A Brief Tale of AstroVirus

Another zoonotic virus with pandemic potential?

Introduction

Quite honestly, I did not know this virus ever existed. I had absolutely no clue about it until recently, when I stumbled across it while I was browsing web for something completely irrelevant. It immediately caught my eyes because of its prefix, “Astro”, the star!!. Admittedly, many virus do have exotic names that are related more or less to their structure and etiology. So, I immediately thought “Astro” might mean the star shaped virus but under the electron microscope at a given magnification (Figure 1), Astrovirus and other non-enveloped icosahedron virus (e.g. SARS) look that much different (at least to me but Astrovirus does look like “star” under the higher magnification). Still, the more I read about this virus, my curiosity grew deeper and deeper. I realized that I was lured to Astrovirus, not by its mystical name that insinuates its secret/mystical features, but by the fact that this virus which has been overlooked for many decades since its discovery could provide important clues about emerging viruses with pandemic potential.

Structure

Astrovirus infects a wide array of mammals, including us, human. A discovery of human Astrovirus was not reported until 1970s by Madeley and Cosgrove who saw 28nm particles under the electron microscopy in stool samples of infants with sever gastroenteritis (*1*).

An etiological agent of the acute gastroenteritis outbreak in infants was named “Astrovirus”, based on the shape of this 28nm particle under the electron microscopy which sort of looked “star” like (Fig. 1A).

The 28nm particle with star like appearance under the electron microscopy (Fig. 1A) is largely due to its novel spike like capsid structure composed of an unique homodimer was indeed the T=3 symmetric icosahederal non-enveloped virus (Fig. 1B), which interestingly resembles those of phylogenically unrelated Hepatitis E virus (2).

A B

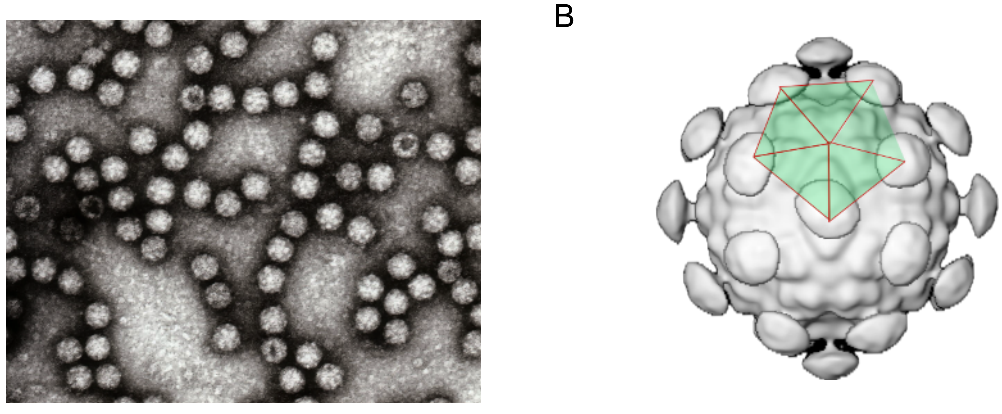

Figure 1 Transmission electron micrograph of Astrovirus particles (A) and a 3D surface reconstructed image of human Astrovirus 1 based on Cryo-EM images, showing T=3 icosahedral symmetry shown by an overlay of green shaded areas with 5 fold symmetric red triangles (added by the author) (B). Sources: (A) Astrovirus Wikipedia (https://en.wikipedia.org/w/index.php?title=Astrovirus&oldid=1311190801) (*3*), and (B) Astroviridae ICTV(https://ictv.global/report_9th/RNApos/Astroviridae) (*4*)

Genome

Genome of Astrovirus is single strand positive sense RNA around 7-8kb in size which is a typical genome size of positive-strand RNA virus except for Coronavirus whose genome is much bigger, somewhere between 27 and 32 Kb (*5*). It is made up of 3 open reading frames (ORFs) ORF 1a and 1b encoding nonstructural inner and outer cores and ORF2 encoding a surface spike protein. The inner and outer core also encodes viral protease and polymerase through translational modification (Fig. 2).

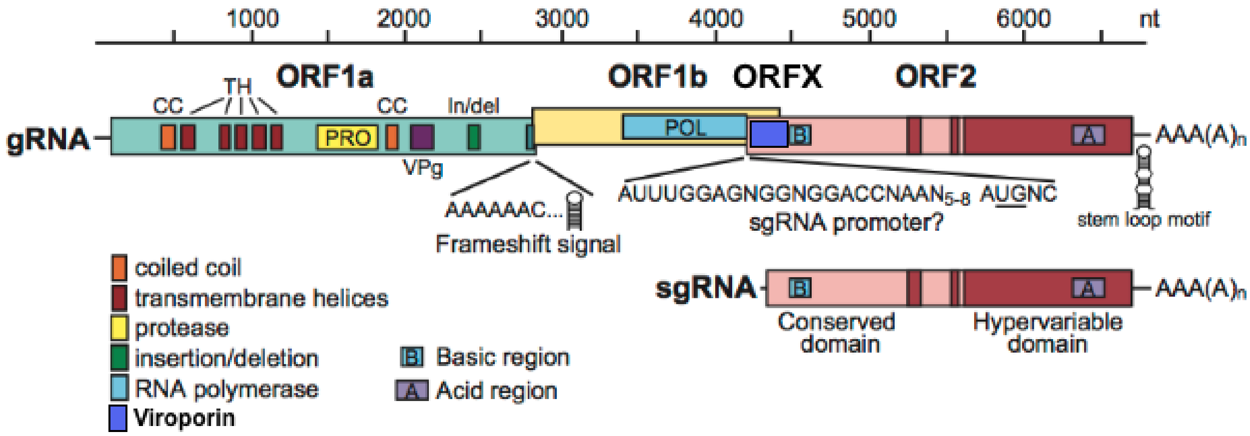

Figure 2 Prototypical genomic organization of Astrovirus shows viral genomic (gRNA) with three ORFs and subgenomic (sgRNA) transcribed from ORF2 of gRNA. ORF1a and 1b encodes nonstructural viral proteins including viral proteases and polymerases while ORF2 encodes structural proteins that form viral capsid. Importantly, ORF2 genomic RNA does not serve as a mRNA for the capsid protein but rather the transcript of ORF2 is generated as sgRNA based on the negative strand template generated during viral replication. Source: Figure 2 from *Astroviridae* ICTV(https://ictv.global/report_9th/RNApos/Astroviridae) (*4*) ORFX reading frame(a dark blue rectangular box) was manually added by the author.

Unlike ORF1, a RNA template for translation of ORF2 gene which encodes structural proteins (viral capsid) is transcribed not through the usual viral genomic RNA but separate smaller subgenomic RNA (sgRNA) (Fig.2). This peculiar process is unique to all the positive-stranded RNA virus in that translation of structural viral proteins must be mediated though transcription of sgRNA (6). Synthesis of sgRNA takes place only after the initiation of viral replication because a template for the transcription of ORF2 sgRNA, a negative strand of gDNA is only generated upon viral replication.

The use of sgRNA, seemingly a highly convoluted approach by Astrovirus (and other positive single-stranded RNA virus) is all about ensuring excess production of viral capsid at the right moment of the viral replication, thereby ensuring optimum viral assembly and efficient virus production. This approach however, only works for the positive single-stranded RNA virus but not for other viruses with distinct replication mechanisms. I guess sgRNA is a brilliant evolutionary ***invention*** for the positive single-stranded RNA virus, however, it is by no means perfect and in fact this mode of viral replication does have many evolutionary pitfalls (7).

There is yet another gene that is worth mentioning. This gene with its fancy name, ORFX, was recently discovered by a non-traditional approach that employs the next generation sequencing (NGS) with modern bioinformatic analysis. ORFX is located near 5' end of ORF2 gene, and encodes a relatively small viral protein with structural and functional resembles to viroporin which enhances a process of viral propagation (i.e. virus budding through the plasma membrane of the infected cell) by increasing plasma membrane permeability of the infected cell (*8*). Although what is discovered here is undoubtedly significant, how ORFX is discovered centrally weights more. As mentioned above, the existence of ORFX was predicted by careful comparative phylogenetic analysis, based solely on NGS across many Astrovirus clades and variants. In addition, the fact that this was done without AI assistance is truly increadible.

Evolution

One of the hallmarks of RNA virus is having an extremely high mutation rate, owing to the fact that replication of RNA genome is mediated by an error prone RNA-dependent RNA polymerase with no proofreading capacity. Astrovirus, being a RNA virus is no exception to this rule, A rate of any single nucleotide substitution by RNA-dependent RNA polymerase is around 3.7×10^{-3} substitutions per site per year, while synonymous substitution (silent mutation) is at about 2.8×10^{-3} per site/year (7, 8). This mutation rate is incredibly high as compared to a typical mutation of DNA virus, for example, Herpes Simplex Virus-1 is around 10^{-8} (*11*).

Perhaps, the most impactful biological process of Astrovirus evolution is genome recombination which actually is a thing of most positive-strand RNA virus. Many positive-strand RNA virus appear to be capable of genome recombination upon co-infection (10). If the virus, as a result of recombination, gains the increased adaptivity and ultimately pathogenicity in human, the results may not be pretty, possibly ending up creating a strain(s) that rapidly causes regional and in the worst case global pandemic like the one (SARS-Cov-2) we just experienced a few years ago.

Within a family of human Astrovirus, evidence of the past recombinational events are embedded in genomes of globally circulating HAstV-3 and 4 (11, 12), and HAstV-8 (13). There has been several examples of novel regional HAstV discovered, and examining genomic sequences of these recombinant HAstVs has revealed a consensus recombination hot spot within ORF1b-ORF2 junction (*16*, *17*).

It should be mentioned that the genomic recombination does not create a brand new de novo virus but rather an antecedent virus with its mosaic genome that is made up of bits and pieces of existing Astrovirus genomes. In the case of HAstVs, it is thought that novel mosaic variants are generated first locally, and then, after multiple rounds of selection, some of them will gain significant increase in the infectivity, and eventually establish global circulation. As of today, there are 8 serotypes of HAstVs circulating globally in human population as well as a few novel HAstVs clades with distinct genotypes found regionally (*18*).

There is yet another interesting twist to the Astrovirus recombination story. The phylogenic studies of Astrovirus have provided evidence suggesting that Astrovirus may share a common ancestal origin with single stranded RNA **plant** virus, *Tymoviridae* and *Potyviviridae.* Astrovirus and *Tymoviridae.* These plant virus do display structural similarity for having T=3 icosahedral capsids. Furthermore, Astrovirus and *Potyviviridae* have the sequence similarity between nonstructural proteins, particularly RNA-dependent RNA polymerase (*16*, *19*, *20*). These viruses most likely split from their common ancestor as a result of the genome recombination in the distant past.

The story of Astrovirus genomic recombination does not stop here at all. Structural resemblance of the capsid among Astrovirus and Hepatitis E virus which bears no phylogenic relation implicates Astrovirus and Hepatitis E virus may be diverged from a common ancestor, which thought to be *Tymoviridae* by series of recombination events (*21*).

Together, these observations strongly suggest that Astrovirus and plant virus share a distant but common ancestor which is most likely archaic plant virus. An intriguing question is whether this archaic plant virus was capable of jumping host species from plant to the vertebrates. Such process is known as a cross-kingdom jump and is rare but has been a few documented cases of plant virus infecting and replicating in insects (*22*, *23*). Evidently, insects are not vertebrates, however it is possible that jumping from plants to insects started an evolutionary process that eventually led to the rise of Astrovirus and perhaps other RNA viruses.

In a nut shell, these data clearly demonstrate that genomic evolution of Astrovirus and perhaps other single stranded RNA viruses appears to be highly mosaic in nature, made up of a combination of bits and pieces of genomic sequences from other related and unrelated single stranded RNA viruses and shuffled them in the prosperous fashion under the persistent evolutionary pressure (this applies to other virus as well). The end results of this whole process is to generate the viable and competent virus that can infect multitude of hosts and jump from one species to another quite readily.

Stem-Loop Motif II

Another one of many interesting genomic features of Astrovirus is **Stem-Loop Motif II** (s2m) found in 3’ untranslated region (UTR), and are found in members of *Caliciviridae*, *Picornaviridae* and *Coronaviridae* (including infamous SARS-CoV-2). The s2m was first discovered in Astrovirus family (19), as a 43 base pair conserved single strand RNA motif at the 3’UTR. While the biological significance of s2m is not fully understood, conservation of s2m in various members of these virus simply indicates its clear contribution to the viral fitness (20). At least *in vitro,* s2m appears to be essential for HAstV-1 viral replication while among SARS-CoV-2, s2m is non-essential for virus life cycle (21). However, observations from in vitro experiments do not always reflect the function of s2m i*n vivo*. In fact some Astrovirus lack s2m such as Melbourne (MLB) clade of HAstVs do exist (*27*, *28*) Despite the lack of s2m, MLB clade is just as prevalent and pathogenic as the most prevalent HAstV1, and most likely will surpass a prevalence rate of the more widespread, HAstV1 sometime in the near future (*29*).

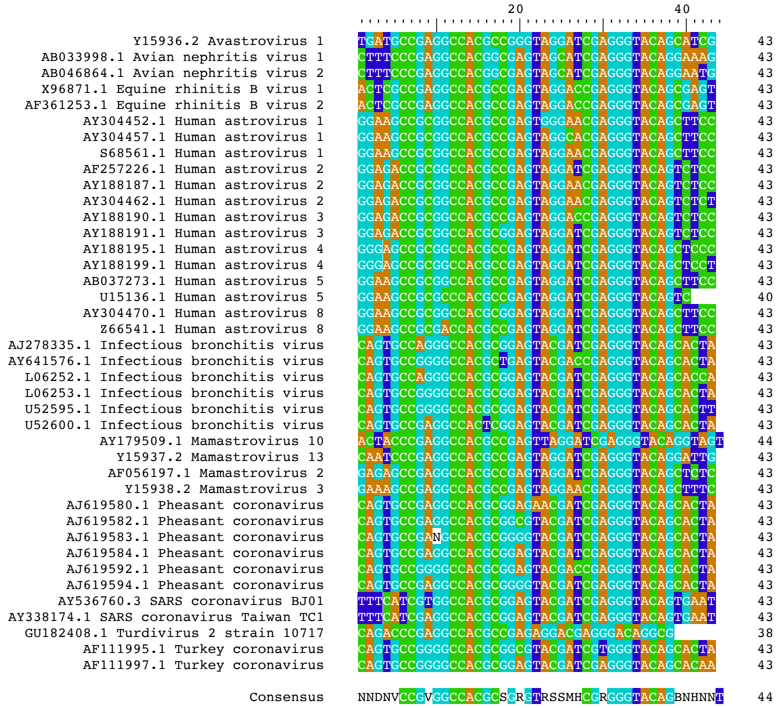

Figure 3 Sequences of s2m from a family of virus known to posses s2m (Family:s2m(RF00164) in Rfam database) are aligned by R package, DECIPER (*30*). This list is by no means exhaustive and there are many other viruses that are unlisted posses the prototypical s2m sequence. A row Left had side shows the NCBI sequence accession number followed by the name of the virus, and the on the right side indicates the length of s2m sequence. A consensus sequence of s2m is shown below the alignment.

Another worth mentioning aspect of s2m is that s2m is found among non-human Astrovirus and Sarbecoviruses with zoonotic potential, particularity in bat may potentially play a role in species jumping. The presence of s2m possibly allows the zoonotic virus to infect a wide range of hosts, including humans. Once the virus establishes itself in the new host/species, s2m may become non-essential and dispensable for virus life cycle in the new host, and most likely end up getting removed from the viral genome, much like a mobile genetic element (20). This notion sounds all good but still largely remains speculation and raise more questions; why is s2m still persisting in the bat Astrovirus and Sarbecoviruses ? and how come s2m is found only in limited members of the Order *Picornavirales* and not widely adapted by other virus ? To address these questions, it is really necessary to grasp biological functions of s2m in the context of Astrovirus natural infection.

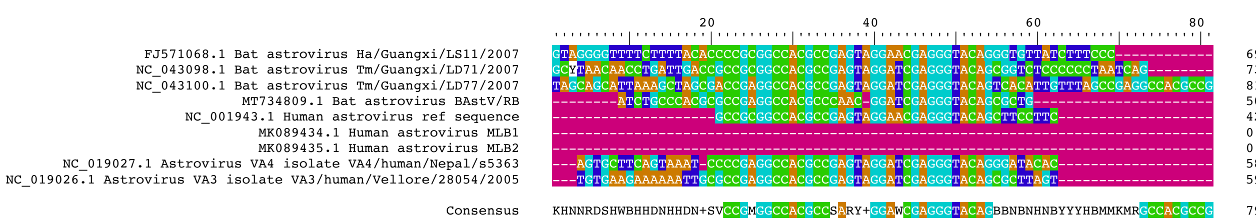

Figure 4 Sequences of s2m from a family of bat and human Astrovirus are aligned by R package, DECIPER (*28*). A row Left had side shows the NCBI sequence accession

number followed by the name of the virus, and the on the right side indicates the length of s2m sequence. A consensus sequence of s2m is shown below the alignment.

Why does current understanding of biological significance of s2m still remain quite obscure despite its decade long characterization with acceptance of its importance in Astrovirusn and Sarbecoviruses pathogenesis?. I think the main issue is how s2m function is characterized in *in vitro* as well as *in vivo* experiments by using s2m deletion mutants. A caveat of using virus with such deletion mutation in general is that they often do not necessarily represent virus exists in wild and thus the infectious model using these mutants barely reflect natural infection.

Moreover, my problem with these experiments in particular is widely held assumption that the manipulation of 3'UTR sequence by either deletion or addition of bit and pieces of RNA will not end up creating artifactual outcomes and with appropriate controls, legitimate results will be obtained. In fact, such manipulation of RNA sequence has considerably a high risk of introducing significant artifactual alterations to the overall secondary structure of the 3'UTR of viral RNA.

Interpretation of results from these studies are not as easy as most biologists/virologists think. Many of my buddy structural biologists often cry wolf about how we, regular bread and butter biologists (i.e. biologist with no structural science training) have the slightest idea about structural consequences of even minor sequence manipulation particularly in RNA (and protein).   Even with thorough controls, it is often difficult to structurally demonstrate preservation of the intact secondary structure after sequence manipulation. Perhaps, the best way to infer actual function of s2m, is to either make a series of progressive s2m deletion as controls, or alternatively, look for s2m deletions in a wild like those recently observed in the SARA-CoV2 Omicron lineage (*31*)

Conclusion

I studied virology/immunology in the graduate school and was certain that I had covered almost all the important viral pathogens for human, and admittedly, I was wrong. I never heard anything about Astrovirus until I stumbled across the existence of Astrovirus recently (roughly 30 years after finishing the grad school). HAstVs including fast emerging variants, clade MLB and VA have been known for the causative agent of pediatric gastroenteritis cases around the globe.

To my knowledge, currently there is no vaccine or antivirals against HAstVs infection, and any development of such treatment options in the near future appears to be given low priority, probably due to the absence of exceedingly sever morbidity and mortality associated with pediatric cases of HAstV infection and also lack of robust academic interest thus insufficient funding available in the HAstV research.

Importantly. Astrovirus share many genomic and physicochemical characteristics as well as pathophysiology, and perhaps, the most important, its zoonotic potential.with other members of the Order *Picornavirales.* Do Astrovirus, particularly bat Astrovirus and emerging new clades of HAstVs, have the potential to cause deadly global pandemic in human ? Probably not because, I think, unlike Coronavirus, Astrovirus in general do not have a big enough genome to be as virulent as Coronavirus (although there is absolutely no correlation between the size of the genome and pandemic potential (Lentivirus genome is around 10kbp)).

Nevertheless, what is the most alarming aspect of Astrovirus is the capability of Astrovirus to undergo intra-genomic (inter clade) and perhaps more dangerous inter-genomic (inter species) recombination. The presence of s2m could possibly be a testimony of how wide spread such recombination in bat and other zoonotic hosts. Are we facing another ticking time bomb of a global scale ? Probably not but we have paid terrible price of underestimating the potential of zoonotic virus not too long ago. Therefore, it is absolutely necessary to have a better understanding of the virology and epidemiology of Astrovirus.

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